Starlink Project Post-Implementation Report

Roscoe D. Govan

Western Governors University

Abstract

I am the IT Manager for our company, and a few months ago, I discovered that our company had been facing severe financial loss annually due to the lack of network redundancy in our store locations. Earlier this year, I proposed a technical solution called Starlink, the newest satellite service provider on the market, to serve as a network failover solution in each store location. The Starlink project was developed based on several key points. These included a needs analysis, cost-benefit analysis, itemized cost breakdown, risk register, qualitative and quantitative risk analysis, timeline, scope statement, milestones, project launch details, implementation strategy, documentation and hardware/software deliverables, project phases, risk mitigation, risk factors, project plan, and evaluation framework. I used three metrics, as previously mentioned, to evaluate the proposed outcome of the project to the actual work of the project, which was "Annual Availability of Services" accompanied by "Power Outages" and "Annual Maintenance Window Time Allotted," which allowed me to come to more accurate conclusions.

Keywords: financial loss, technical solution, key points, three metrics

Table of Contents

Abstract 2

Quality Assurance 6

Formative Evaluation and QA Metrics 6

Testing Methodology 8

Test Cases .....**Error! Bookmark not defined.**

Acceptance Criteria 11

Project Review 12

Assumptions 13

Project Phases 13

Timeline Deviations 18

Project Dependencies 18

Resource Requirements 19

Implementation Project Milestones 20

Project Deliverables 23

Documentation Deliverables 23

Formative Evaluation Results and Revisions 33

Summative Evaluation Plan and Results 35

Stakeholder Communication Plan and Reports 35

Ongoing Support and Maintenance 40

Resources for Post-Implementation Support.....40

Short and Long Term Maintenance Plan 42

Post-Implementation Project Summary.....43

Documentation Deliverables.....43

Success Criteria for Each Project Outcome 50

Justification of Proposed versus Actual Project Outcomes.....52

Lessons Learned 52

References 54

Tables 56

Post Implementation Report Title

I am the IT Manager for our company, and a few months ago, I discovered that our company had been facing severe financial loss annually due to the lack of network redundancy in our store locations. Each of the company stores was relying on a single physical cable to facilitate all internet connectivity. This means that if the cable was damaged, either inside or outside the store location, the entire store would be rendered offline. My investigation revealed that our company only had 83% availability annually, while the current industry standard is 99.999%. After calculating the financial loss, we incurred due to network redundancy, I found that it amounted to four million dollars in the previous year, in addition to the unacceptable percentage of availability. After analyzing the situation, I gained a clear understanding of our organization's current state. I then began searching for a solution to the identified problem.

Starlink's unique feature is that its satellites are positioned closer to Earth. So, compared to other providers such as HughesNet or ViaSat, the signal does not have to travel as far, resulting in significantly lower latency and higher service quality. I discovered that Starlink provides four business plans, and based on our needs, the package we required for our stores was much more affordable than HughesNet or ViaSat. Lastly, I learned that SpaceX, backed by billionaire Elon Musk, introduced Starlink in 2019 and that their goal is to scale the product to provide global coverage. Based on these factors, I knew I had found the best solution for our business, and the only thing left to do was to take the data and turn it into information to be used as business intelligence for you as leadership to make the right decision. Based on these factors, I found the best solution for our business. The only thing left to do was to take the data and turn it into information to be used as business intelligence for you as leadership to make the right decision.

In my original proposal I included all of the aspects that I knew would be necessary for you to make a educated decision regarding whether to approve the project or deny the project. My proposal had several key points that I was confident would help with the decision to include a needs analysis, cost-benefit analysis, itemized cost breakdown, risk register, qualitative and quantitative risk analysis, timeline, scope statement, milestones, project launch details, implementation strategy, documentation and hardware/software deliverables, project phases, risk mitigation, risk factors, project plan, and evaluation framework. I submitted the proposal to leadership in January of 2024 with an expected start date of April 1, 2024. I knew it would take time for finances to be assessed, meetings to be held, questions that may arise and more, so I allotted approximately two and a half months before project start to get all those aspects out of the way. The project ended up being approved and was a massive success in the end. I would like to personally thank each one of you for your time, wisdom, and collaboration on this incredible venture that is going to take us one step closer to being the company of the future!

Quality Assurance

As a project manager for the Starlink project, I utilized various tools and techniques to ensure quality management. Specifically, I employed the "Audits, Data Gathering, and Data Analysis" techniques. Our project team began the installation process on May 1, 2024. In May, I conducted in-person quality assurance assessments for 75 installations, while for the remaining 125 installations, we relied on archived photos of each part of the installation to evaluate them when time was allotted. To ensure that the installations were executed as expected, I requested the network architects to have the telecommunications technician who installed the hardware send clear photos of all installation components. I provided clear examples of what the pictures needed

to show and the distance the pictures should be taken from. The network architects captured and archived photos for each site for me to evaluate for potential rework. I performed the same process during the sprint that took place in June, and July. I used a form to assess every site, regardless of whether it was evaluated in person or via the archived photos. The audit included the following questions:

1. Is there anything visible that could lead to damage or a blocked signal? (Y/N)
    - If yes, have we explored alternative locations for the equipment?
    - If yes, how likely is it for the potential damage to happen? *(Est. Percentage Here)*
2. Is there any visible damage to the Starlink equipment? (Y/N)
    - If yes, what components are damaged?
    - If yes, is the damage internal or external or both?
    - If yes, what are the network statistics showing in the Starlink portal?
3. Are all the expected components present in the setup? (Y/N)
    - If "No," which components are missing?
    - If "No," what is the technicians reasoning?
4. Is the satellite pointing towards space? (Y/N)
    - If "No," can it be remotely power cycled to adjust itself?
5. Does the Starlink portal show that the service is online? (Y/N)
    - If "No," what is the telecommunications technicians reasoning?
    - If "No," what is the network architects reasoning?
6. Does the Starlink portal show an obstruction above 0.02%? (Y/N)
    - If yes, what is the percentage?

7. Is the correct serial number displayed for the site in the Starlink portal? (Y/N)
    - If “No,” what is the network architects reasoning?
    - If “No” does Starlink come online when the right SN is entered?

I waited until after hours daily to test the implementations. I remotely accessed the sites and manually failed them to Starlink to conclude the implementation's success and confirm the network architects' reports. I left the stores on Starlink for five days for data-gathering purposes. I observed the amount of traffic the stores had when on Starlink and how well it performed. I then performed data analysis and decided whether the Starlink priority 1TB business plan was necessary or if one or more stores needed to be dropped down to the Starlink standard business plan. I did this to ensure we were financially responsible throughout the project. I have archived the form, photos, and evaluation results to utilize them in the future to benefit our company. This data can be tracked to support decision-making based on evidence in upcoming projects. It will help us make informed decisions when we encounter a difficult situation.

Formative Evaluation and QA Metrics

I made project scheduling one of my top priorities during the Starlink project. To keep the project on track, I utilized the Pert & Gant chart. I shared this chart with the project team and stakeholders in a read-only format, so everyone was in the loop regarding the project's progress. By using this approach, we could identify any potential issues ahead of time, and stakeholders could reach out to me with any questions or concerns. We relied on the project milestones to track the project's progress, and these milestones were assigned a percentage of completion. The key milestones that we tracked were:

1. Project Charter Completed - Project 12.5% Complete

2. Starlink Equipment Received - Project 25% Complete
3. Project Team Trained - Project 37.5% Complete
4. 200 Installations Complete - Project 50% Complete
5. 400 Installations Complete - Project 62.5% Complete
6. 500 Installations Complete - Project 75% Complete
7. Project Deliverables Accepted - Project 87.5% Complete
8. Project Closure Tasks Complete – Project 100% Complete

To ensure that the project progressed smoothly, I also utilized checklists which allocated a percentage towards each task that amounted to 100%. If all aspects of a Starlink installation met expectations, the checklist was marked as 100% complete. An example of the checklists used during this project has been provided below for reference.

Telecommunications Technician Checklist

Enter Store Number Here:

Date	Task Completed	Issues Encountered	Reported Issue	Issue Explanation	Task Complete/Not Complete
##/##/####	Installed Starlink Hardware Dish?	Y/N	Y/N		
##/##/####	Confirmed Starlink Hardware Dish is pointed towards space?	Y/N	Y/N		
##/##/####	Installed power cord that goes from Starlink Dish to Power Supply?	Y/N	Y/N		
##/##/####	Confirmed no damage to the power cord that goes from Starlink Dish to Power Supply?	Y/N	Y/N		
##/##/####	Installed Starlink Power Supply in a well ventilated area?	Y/N	Y/N		
##/##/####	Confirmed Starlink Power Supply indicator light showed green?	Y/N	Y/N		
##/##/####	Installed the Starlink Power Supply power cord to working outlet?	Y/N	Y/N		
##/##/####	Confirmed Starlink Proprietary Cable was functional and connected to network switch?	Y/N	Y/N		
##/##/####	Worked with Network Architect until Starlink was confirmed online?	Y/N	Y/N		
##/##/####	Sent clear photos of each Starlink component after install?	Y/N	Y/N		

Network Architect Checklist

Enter Store Number Here:

Date	Task Completed	Issues Encountered	Reported Issue	Issue Explanation	Task Complete/Not Complete
##/##/####	Validated Starlink Hardware Dish?	Y/N	Y/N		
##/##/####	Confirmed Starlink Hardware Dish is pointed towards space?	Y/N	Y/N		
##/##/####	Validated power cord that goes from Starlink Dish to Power Supply?	Y/N	Y/N		
##/##/####	Confirmed no damage to the power cord that goes from Starlink Dish to Power Supply?	Y/N	Y/N		
##/##/####	Validated Starlink Power Supply in a well ventilated area?	Y/N	Y/N		
##/##/####	Confirmed Starlink Power Supply indicator light showed green?	Y/N	Y/N		
##/##/####	Validated the Starlink Power Supply power cord to working outlet?	Y/N	Y/N		
##/##/####	Confirmed Starlink Proprietary Cable was functional and connected to network switch?	Y/N	Y/N		
##/##/####	Worked with Telecommunication Technician until Starlink was confirmed online?	Y/N	Y/N		
##/##/####	Archived clear photos of each Starlink component after install?	Y/N	Y/N		
##/##/####	Configured the Starlink equipment to work as a failover for the store?	Y/N	Y/N		
##/##/####	Compared configuration to a known working site?	Y/N	Y/N		
##/##/####	Updated knowledge base with anything new?	Y/N	Y/N		
##/##/####	Educated & Involved network administrators?	Y/N	Y/N		
##/##/####	Checked Starlink Portal for SN accuracy?	Y/N	Y/N		
##/##/####	Checked Starlink Portal to ensure no obstruction present?	Y/N	Y/N		

(I divided the total amount of tasks in a checklist by 100 to attain the value of each task. For example, if ten tasks were on the list, each task would be worth 10%.)

Testing Methodology

We had installed Starlink as a backup internet solution for our company's store locations. To ensure that the installation was successful, we needed to confirm that the equipment worked correctly by manually failing over services at each store location. To test the Starlink equipment, I accessed it remotely during the night and shut down the primary internet connection to determine whether Starlink could take over. Since no customers were shopping at night, I left the direct fiber link down for up to two days to analyze Starlink's ability to handle the amount of data each store transmitted. This helped me determine whether each store needed the standard business plan, or the premium business plan offered by Starlink. After two days, I manually reconnected the primary fiber connection and confirmed that everything returned to normal at the stores. I documented all the findings in an Excel file, including the outcomes and any actions needed. I archived these documents for each store to ensure that there was no question about whether the functionality of the Starlink solution was tested. The electronic records also included screenshots to provide further evidence of the results.

Test Cases

Test Cases							
Store#	**Business Plan**	**Test**	**Start Time**	**End Time**	**Success/Fail**	**Action Plan *(If needed)***	**Feedback**
1	Priority 1TB	Starlink Failover	5/6/2024	5/8/2024	Success	No additional action needed	The store successfully switched to using Starlink as its primary connection when the fiber connection failed. Once the fiber connection was restored, it took back over as the primary connection. During the time when Starlink was in charge, it handled the store's traffic well. The equipment's firmware update did not cause any functional issues.
		Primary Fiber Recovery	5/6/2024	5/8/2024	Success		
		Starlink Data Handling	5/6/2024	5/8/2024	Success		
		Starlink Service	5/6/2024	5/8/2024	Success		
		Starlink Firmware Update	5/6/2024	5/8/2024	Success		
2	Priority 1TB	Starlink Failover	5/6/2024	5/8/2024	Success	No additional action needed	The store successfully switched to using Starlink as its primary connection when the fiber connection failed. Once the fiber connection was restored, it took back over as the primary connection. During the time when Starlink was in charge, it handled the store's traffic well. The equipment's firmware update did not cause any functional issues.
		Primary Fiber Recovery	5/6/2024	5/8/2024	Success		
		Starlink Data Handling	5/6/2024	5/8/2024	Success		
		Starlink Service	5/6/2024	5/8/2024	Success		
		Starlink Firmware Update	5/6/2024	5/8/2024	Success		
3	Priority 1TB	Starlink Failover	5/6/2024	5/8/2024	Success	No additional action needed	The store successfully switched to using Starlink as its primary connection when the fiber connection failed. Once the fiber connection was restored, it took back over as the primary connection. During the time when Starlink was in charge, it handled the store's traffic well. The equipment's firmware update did not cause any functional issues.
		Primary Fiber Recovery	5/6/2024	5/8/2024	Success		
		Starlink Data Handling	5/6/2024	5/8/2024	Success		
		Starlink Service	5/6/2024	5/8/2024	Success		
		Starlink Firmware Update	5/6/2024	5/8/2024	Success		
4	Priority 1TB	Starlink Failover	5/6/2024	5/8/2024	Success	No additional action needed	The store successfully switched to using Starlink as its primary connection when the fiber connection failed. Once the fiber connection was restored, it took back over as the primary connection. During the time when Starlink was in charge, it handled the store's traffic well. The equipment's firmware update did not cause any functional issues.
		Primary Fiber Recovery	5/6/2024	5/8/2024	Success		
		Starlink Data Handling	5/6/2024	5/8/2024	Success		
		Starlink Service	5/6/2024	5/8/2024	Success		
		Starlink Firmware Update	5/6/2024	5/8/2024	Success		
5	Priority 1TB	Starlink Failover	5/6/2024	5/8/2024	Success	No additional action needed	The store successfully switched to using Starlink as its primary connection when the fiber connection failed. Once the fiber connection was restored, it took back over as the primary connection. During the time when Starlink was in charge, it handled the store's traffic well. The equipment's firmware update did not cause any functional issues.
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		Starlink Data Handling	5/6/2024	5/8/2024	Success		
		Starlink Service	5/6/2024	5/8/2024	Success		
		Starlink Firmware Update	5/6/2024	5/8/2024	Success		
6	Priority 1TB	Starlink Failover	5/6/2024	5/8/2024	Success	No additional action needed	The store successfully switched to using Starlink as its primary connection when the fiber connection failed. Once the fiber connection was restored, it took back over as the primary connection. During the time when Starlink was in charge, it handled the store's traffic well. The equipment's firmware update did not cause any functional issues.
		Primary Fiber Recovery	5/6/2024	5/8/2024	Success		
		Starlink Data Handling	5/6/2024	5/8/2024	Success		
		Starlink Service	5/6/2024	5/8/2024	Success		
		Starlink Firmware Update	5/6/2024	5/8/2024	Success		
7	Priority 1TB	Starlink Failover	5/6/2024	5/8/2024	Success	No additional action needed	The store successfully switched to using Starlink as its primary connection when the fiber connection failed. Once the fiber connection was restored, it took back over as the primary connection. During the time when Starlink was in charge, it handled the store's traffic well. The equipment's firmware update did not cause any functional issues.
		Primary Fiber Recovery	5/6/2024	5/8/2024	Success		
		Starlink Data Handling	5/6/2024	5/8/2024	Success		
		Starlink Service	5/6/2024	5/8/2024	Success		
		Starlink Firmware Update	5/6/2024	5/8/2024	Success		

The Starlink project was completed in a series of 30-day work periods, known as sprints. During each sprint, we installed ten units per week and performed a test method on every store that was operational. I have included a visual of some test cases I completed. In total, there were 500 test cases, all of which have been electronically archived for future reference.

Each test case was documented with the date and time of initiation and conclusion, whether it passed or failed, provided a section for corrective actions to be outlined, and feedback explaining why it was a success or failure. With so many daily, weekly, and monthly tasks involved in a project, it is important to simplify implementations as much as possible to maximize efficiency.

This test method was simple and effective, allowing for quick analysis of the installations, which ultimately led to the acceptance of the deliverables.

Acceptance Criteria

As part of the final evaluation process for the Starlink project, I validated that every store where we installed the Starlink solution was still working perfectly, both in terms of hardware and software. During the execution period, I conducted quality assessments for all 500 locations, and reviewed all the installations for that sprint. The final review was conducted systematically by me, network architects, and administrators. We logged into each store late at night before scope validation and confirmed that the Starlink solution was still functioning as expected. We ensured that every store successfully operated when manually failing over to the Starlink solution, and there was no visible impact on service quality. We also confirmed that all industry standards and regulations, such as PCI DSS and HIPA, were not affected regarding our deployment of the Starlink solution. Doing all this ensured that the deliverables were guaranteed to be accepted as soon as possible for scope validation, as the terms for closure were that all project stakeholders had to sign off on both the hardware deployments and software functionality at all our 500 store locations.

Project Review

In January 2024, I discovered that our company had been facing severe financial loss annually due to the lack of network redundancy in our store locations. Each company store relied on a single physical cable to facilitate all internet connectivity. This means that if the line was damaged, inside or outside the store location, the entire store would be rendered offline. My investigation revealed that our company only had 83% availability annually, while the current industry standard is 99.999%. After calculating the financial loss we incurred due to network

redundancy, I found that it amounted to four million dollars in the previous year, in addition to the unacceptable availability percentage. After analyzing the situation, I clearly understood our organization's current state. I then began searching for a solution to the identified problem.

Starlink's unique feature is that its satellites are positioned closer to Earth. So, compared to other providers such as HughesNet or ViaSat, the signal does not have to travel as far, resulting in significantly lower latency and higher service quality. I discovered that Starlink provides four business plans, and based on our needs, the package we required for our stores was much more affordable than HughesNet or ViaSat. Lastly, I learned that SpaceX, backed by billionaire Elon Musk, introduced Starlink in 2019 and that their goal is to scale the product to provide global coverage. Based on these factors, I knew I had found the best solution for our business, and the only thing left to do was to take the data and turn it into information to be used as business intelligence for you as leadership to make the right decision. Based on these factors, I found the best solution for our business. The only thing left to do was to take the data and turn it into information to be used as business intelligence for you as leadership to make the right decision. The only thing left to do was to take the data and turn it into information to be used as business intelligence for you as leadership to make the right decision. After presenting the necessary information to company leadership as a formal proposal for “The Starlink Project,” we completed the Starlink project ahead of the planned schedule and are now at the point of receiving the return on investment.

Assumptions

In my proposal for the Starlink project, I outlined five assumptions. Firstly, I assumed that we could hire the necessary personnel for a salary that is lower than the national median

average displayed by the Federal Bureau of Labor Statistics. Secondly, I assumed that Starlink could fulfill an order of 600 hardware kits simultaneously without requiring any additional charges. Thirdly, I assumed that our company had sufficient funds to finance the Starlink project. Fourthly, I assumed that all the stakeholders involved at the beginning of the project would remain with the company until the end of the project. Lastly, I assumed that customers would notice the positive changes and adapt their overall perspective of the company accordingly. I successfully validated all my assumptions. I skillfully hired top talent for less than the national median average, ordered 600 Starlink hardware kits in a single go, got the Starlink project and its budget approved, and masterfully influenced the customer's perspective to gain unwavering positive regard for the company. All of my assumptions turned out to be correct. I was able to hire the necessary talent for less than the national median average, order 600 Starlink hardware kits at once, get the approval of the Starlink project and its budget and sway the customer's perspective enough to make the company begin to receive positive regard.

Project Phases

Timeline Visual:

ID	Task	Phase/Process	Duration	Start	Finish
1	Starlink Project Official Launch	Project Charter Completion - Project Launch	1 day	4/1/2024 8:00	4/1/2024 17:00
2	Call Starlink To Discuss Order	Project Management Execute Process Group	1 day	4/2/2024 8:00	4/2/2024 17:00
3	Starlink Equipment Gets Ordered	Project Management Execute Process Group	1 day	4/3/2024 8:00	4/3/2024 17:00
4	Starlink Equipment Arrives	Project Management Execute Process Group	6 days	4/15/2024 8:00	4/20/2024 17:00
5	Hire Project Team	Project Management Execute Process Group	12 days	4/1/2024 8:00	4/12/2024 17:00
6	Establish 24 Processes In PM Planning Process Group	Project Planning & Design Completion	29 days	4/2/2024 8:00	4/30/2024 17:00
7	Project Team Members Are Trained	Project Management Execute Process Group	16 days	4/15/2024 8:00	4/30/2024 17:00
8	Scrum Planning Meeting	Scrum Planning Meeting Meeting	1 day	5/1/2024 8:00	5/1/2024 17:00
9	Discuss Roadmap Of Sprint W/ Team	Scrum Initiation Phase	1 day	5/1/2024 8:00	5/1/2024 17:00
10	Plan The Current Sprint Details	Scrum Planning Phase	1 day	5/1/2024 8:00	5/1/2024 17:00
11	Installation Audits	Quality Review 1	10 days	5/15/2024 8:00	5/24/2024 17:00
12	Installations Begin	Scrum Execution Phase	30 days	5/1/2024 8:00	5/30/2024 17:00
13	Scrum Sprint Review Meeting	Scrum Review Phase	1 day	5/31/2024 8:00	5/31/2024 17:00
14	200 Installs Complete Milestone	MileStone - 200 Installations Complete	31 days	5/31/2024 8:00	6/30/2024 17:00
15	Scrum Planning Meeting	Scrum Planning Phase	1 day	6/1/2024 8:00	6/1/2024 17:00
16	Installation Audits	Quality Review 2	10 days	6/15/2024 8:00	6/24/2024 17:00
17	Installations Persist	Scrum Execution Phase	30 days	6/1/2024 8:00	6/30/2024 17:00
18	Scrum Sprint Review Meeting	Scrum Review 2	2 days	6/30/2024 8:00	7/1/2024 17:00
19	400 Installs Complete Milestone	MileStone - 400 Installations Complete	30 days	6/1/2024 8:00	6/30/2024 17:00
20	Plan The Current Sprint Details	Scrum Planning Phase	1 day	7/1/2024 8:00	7/1/2024 17:00
21	Quality Assessment	Quality Review 3	15 days	7/1/2024 8:00	7/15/2024 17:00
22	Installations Persist	Scrum Execution Phase	15 days	7/1/2024 8:00	7/15/2024 17:00
23	Scrum Sprint Review Meeting	Scrum Review 3	1 day	12/9/2023 8:00	12/9/2023 17:00
24	500 Installs Complete Milestone	MileStone - 500 Installations Complete	15 days	6/1/2024 8:00	6/15/2024 17:00
25	Scope Validation	Project Deliverable Acceptance	5 days	7/16/2024 8:00	7/20/2024 17:00
26	Room For Rework	Rework	44 days	7/20/2024 8:00	9/1/2024 17:00
27	Closure Tasks	Project Closure	30 days	9/1/2024 8:00	9/30/2024 17:00

During phase one of the Starlink project I obtained the official project charter authorizing me to be the central authority over the Starlink project and have the project officially recognized by the organization. Additionally, in this phase I formally identified all stakeholders that were to be involved in the project, and successfully hired all 15 necessary project team members for training to begin as expected on 4/15/2023. During this phase I went ahead and ordered the 600 Starlink hardware kits as I understood it would take 1-2 weeks for the equipment to arrive. I was able to successfully accomplish all these tasks by 4/15/2024, is in alignment with the expected project timeline.

During phase two of the project, I worked with the stakeholders and project team to get through all 24 processes in the planning process group of the PMBOK framework. It was during this phase answered questions as to how many people would be involved in each installation, what requirements stakeholders would require, formally solidifying the scope statement to all involved in the project. We also received the Starlink equipment as of 4/17/2024, which worked out perfectly regarding the project schedule as we were able to get the new hires trained and ready to work by 4/31/2024. Our Scrum planning meeting held on 5/1/2024 was the transition point from phase two to phase three.

We entered phase three, the execution phase, on 5/1/2024. Upon conclusion of our sprint planning meeting, where we discussed the roadmap of the project with the team and elaborated on the current sprint day and month, the project team was immediately dispatched out to their predetermined locations to begin installing the hardware and working with the network architects to configure the software the test failover on the network. The team successfully installed the expected 200 Starlink hardware equipment packages by 5/31/2024. The quality assessments that

I performed sporadically from May 1, 2024, to May15, 2024, showed that the installations were all done impeccably. Between 5/2/2024 and 5/17/2024, I noticed that the team had surmounted the learning curve, and though we were not behind schedule, installations were progressing much more fluidly. To close the current sprint and transition to the next, we held a retrospective meeting between the project team and you as stakeholders to review the work that had been performed within this sprint. The meeting was short but went very well, as it was documented in our notes that all stakeholders were impressed with the swift progression of the project. As of this point, we had hit our first milestone in the project and communicated the progress to all stakeholders to build excitement and continue improving communication.

We continued in the execution phase on June 1, 2024, repeating the same process we did on May 1, 2024. We held another sprint planning meeting on June 1/2024, where we discussed the roadmap of the project with the team and elaborated on the current sprint day and month; the project team was immediately dispatched out to their predetermined locations to begin installing the hardware and working with the network architects to configure the software the test failover on the network. The team successfully reached our goal of 400 total Starlink hardware equipment installations by 6/31/2024. The quality assessments that I performed sporadically from June 1, 2024, to June 10, 2024, showed that the installations were all done impeccably. To close the current sprint and transition to the next, we held a retrospective meeting between the project team and you as stakeholders to review the work that had been performed within this sprint. The meeting was short but went very well, as it was documented in our notes that all stakeholders were still impressed with the progression and execution of the project. During the second sprint the project team demonstrated high levels of efficiency as they completed installations fast enough to analyze the quality of their work before leaving the site. As of this

point, we had hit our second big milestone in the project and once again communicated the progress to all stakeholders to build excitement and continue improving communication.

We continued in the execution phase on July 1, 2024, repeating the same process we did in both May and June. However, we only had 100 Starlink hardware kits left to install, so the end of the 3rd sprint was expected to be half the time of the previous. We held another sprint planning meeting on July 1/2024, where we discussed the roadmap of the project with the team and elaborated on the current sprint day and month; the project team was immediately dispatched out to their predetermined locations to begin installing the hardware and working with the network architects to configure the software and test failover on the network. The project team successfully completed the 100 remaining installations as of July 15, 2024, as expected. The quality assessments that I performed sporadically from July 1, 2024, to July 15, 2024, showed that the installations were all done impeccably. To close the current sprint and transition to the next phase, we held a retrospective meeting between the project team and you as stakeholders to review the work that had been performed within this sprint. The meeting was short but went very well, as it was documented in our notes that all stakeholders were still impressed with the progression and execution of the project. As of this point, we had hit our final big milestone of 500 installations completed in the project and once again communicated the progress to all stakeholders to keep excitement and persist in communication. The fourth phase of the Starlink project was monitoring and controlling which ran concurrent with the project execution. The quality assessments that were performed during every sprint are representative of monitoring and controlling being integrated into the Starlink project.

I wanted to make sure that there was time remaining for potential rework that could be presented when we went about scope validation, so July 16, 2024 – Sept 15, 2024, was left open

for potential project schedule delays, or rework. However, we moved towards scope validation as of July 16, 2024 – July 20, 2024, and all of you as project stakeholders signed off on one the deliverables of the project as of July 20, 2024, which marked a day of amazing accomplishment for our organization. Since the deliverables were accepted, we were able to complete all project closure requirements by July 31, 2024, instead of September 31, 2024.

Timeline Deviations

There was only one deviation in the project timeline. We finished the project ahead of time. Initially, we had proposed a completion date of September 31, 2024. However, due to the work being done precisely as planned and you all signing off on all deliverables on July 20, 2024, we proceeded to project closure much faster than expected without needing rework. We completed the entire project an entire month ahead of time. The official project close date was back on July 31, 2024.

Project Dependencies

Dependency #1 - To ensure the successful completion of the Starlink project, it was crucial to identify and address any dependencies in the timeline. This helped to avoid any complications that would have arisen after the project had been approved. This project had five dependencies in the Starlink project that need clarification.

Dependency #2 - To initiate a project and grant me the authority to oversee it, I had to wait to receive the project charter officially. All other activities in the project could only begin once this was complete. This document marked the official commencement of the project. After the project charter, the following dependency was regarding hiring the new team members by April 15, 2024, to offer two weeks of training and begin working by May 1, 2024.

Dependency #3 - Another dependency we faced was regarding technicians being able to move on to their next installation. Technicians could only move to the next job once they have successfully worked with the network architect to get Starlink online at the location they are working on or if management has advised that they can do so based on more pressing project parameters.

Dependency #4 - You as stakeholders had to approve the deliverables before we could officially close the project. Only some people could give the sign-off, so if they could prove that the deliverables we produced were outside the initial scope, we will be subject to either rework or the project being deemed a failure and facing cancellation where the sunk costs are accepted.

Resource Requirements

The resources required for the Starlink project included six hundred Starlink hardware kits, five hundred service activation requests, five network architects, three network administrators, seven telecommunications technicians, and one junior project manager. I was able to procure all the necessary personnel, order all the equipment needed for the project, and get our five hundred stores activated with Starlink's 1TB service package.

The six hundred Starlink hardware kits was an investment of $1,500,000. At a minimum we only needed five hundred of these kits, but I requested funds to secure additional to prevent delays in case out of box failure were experienced during the execution phase of our project. Though we did not use the additional because none of the equipment ended up being inoperable upon arrival, we now have additional equipment that will support the swift replacement of malfunctioning components in the future for our store locations. As of May 1, 2024, the project team had been tasked with the installation of ten Starlink hardware kits per week. The team completed two hundred installations back on May 31, 2024. The project team completed another

200 installs back on June 31, 2024. The project team finished the last 100 installations on July 15, 2024. Starlink service was activated and validated at every store location by July 15, 2024.

During the project execution phase, the telecommunications technicians installed Starlink hardware across our five hundred store locations. The technician at the store location contacted one of the five network architects to configure the equipment as a failover solution. The network administrators monitored for any alerts indicating that a store, for which we had confirmed Starlink as a failover either failed over or required our attention due to unexpected failure. All the project participants worked exceptionally well together, leading to project success. The junior project manager gained valuable experience but was not required to take over the Starlink project, as there were no other projects launched concurrent to the Starlink project. The junior project manager can now be even more knowledgeable when taking on future projects for the company. Lastly, according to the Federal Bureau of Labor Statistics, I successfully saved the company money by procuring all the necessary human resources at less than what showed to be the national average median pay for each of the positions advertised.

Implementation Project Milestones

- Milestone 1 - Project Charter Completion – *Started: April 1, 2024 – Ended: April 1, 2024*

April 1, 2024, was marked as the first official day of the Starlink project as it was also the date that I received the signed project charter, and formally identified all stakeholders in the project.

- Milestone 2 - Project Planning Completion – *Started: April 1, 2024 – Ended: April 31, 2024*

From April 1, 2024 – April 31, 2024, I worked with the project team and stakeholders to complete the 24 processes that are found in the PMBOK project management framework planning process group.

- Milestone 3 - Project Design Completion – *Started: April 1, 2024 – Ended: April 31, 2024*

From April 1, 2024 – April 31, 2024, while planning the project, I also worked concurrently with the project team and stakeholders to formally agree upon the visual of the current state of our company versus the desired state we were working towards having by September 31, 2024.

- Milestone 4 - Project Team Hired - *Started: April 1, 2024 – Ended: April 15, 2024*

Hiring advertisements went out for the necessary technicians as of April 1, 2024, and by April 15, 2024, I had successfully interviewed enough applicants to make a decision on the 15 new hires for the project.

- Milestone 5 - Project Team Trained – *Started: April 15, 2024 – Ended: April 31, 2024*

The new hires started training as of April 15, 2024, and were done with training and ready to begin working as of April 31, 2024.

- Milestone 6 - Starlink Equipment Arrives – Started: *April 1, 2024 – Ended: April 17 - 2024*

I placed the order for the six hundred Starlink hardware kits as of April 1, 2024, because I knew that Starlink had a 1-2 week waiting time for orders. The six hundred Starlink hardware kits arrived as of April 17, 2024. I had projected the arrival between April 15, 2024, and April 20, 2024, so the ordering of equipment went as planned.

- Milestone 7 - 200 Installations Complete – *Started: May 1, 2024 – Ended: May 31, 2024*

We began the first 30-day sprint with the goal of completing 200 initial installations of Starlink hardware on May 1, 2024. The project team successfully completed the 200 installations back on May 31, 2024.

- Milestone 8 - 400 Installations Complete – *Started: June 1, 2024 – Ended: June 30, 2024)*

We began the second 30-day sprint with the goal of completing an additional 200 installations back on June 1, 2024. The project team successfully completed the 200 installations, which brought us to a total of 400 successful installations back on June 30,2024

- Milestone 9 - 500 Installations Complete – *Started: July 1, 2024 – Ended: July 15, 2024*

We began the third sprint back on July 1, 2024, knowing that it would not be a 30-day sprint, due to the number of installations we had left to meet holistic the project team successfully completed the 100 installations, which brought us to a total of 500 successful installations back on July 15, 2024.

- Milestone 10 - Project Deliverables Accepted *(July 16, 2024 – July 20, 2024)*

We completed all installations as planned back on July 15, 2024. So, we moved towards scope validation on July 16, 2024, where we actively engaged you as key stakeholders and got the formal acceptance of all project deliverables back on July 20, 2024.

- Milestone 11 - Project Closure – *Started: July 21, 2024 – Ended: July 31, 2024*

Since we had successfully gotten approval of the project deliverables and did not have to do any rework, we moved to the project closure tasks and were able to complete the project before September 31, 2024. The project was officially complete back on July 31, 2024.

Project Deliverables

In the Starlink project, we installed 500 Starlink hardware kits. Each kit included 1 Starlink Satellite Dish, 2 Starlink Power Cables, 1 Starlink Power Supply, and 1 Starlink Proprietary Ethernet Cable. I requested an extra 100 Starlink hardware kits as a precautionary measure in case any of the kits were damaged during delivery. In the end, I was able to deliver all 500 kits successfully and kept the additional 100 kits in storage for future use by the company. The commitment to providing Starlink service to all 500 store locations was made in addition to the hardware agreement. On July 31, 2024, you approved the Starlink hardware and software in stores by signing off on the scope validation checklist. The following documents were delivered at the end of the project: Risk Register, Stakeholders Communications Plan, Project Schedule, Assumptions Log, Documentation Deliverables, Qualitative Analysis Chart, Quantitative Analysis Chart, and the Project Plan.

Documentation Deliverables

Below I have provided graphic evidence that we did indeed install Starlink. I have included a visual display of each of the components that went into place to get the service up and running for us. Graphic #1 is a visual display of one Starlink deployment at store number eighty-five. Graphic #2 is a visual display of the nuts and bolts that came with the Starlink hardware kits the project team installed. Graphic #3 is a visual display of the Starlink Dish and a first look at the power supply our hardware kits came with. We installed these power supplies, which power the Starlink satellite dish, at every store. Graphic #4 is a visual of our new Starlink software as a dashboard that allows us to track how well Starlink operates when the primary fiber connection goes down for any reason. This software is centralized software that will enable us to monitor the

service quality of Starlink flowing through all 500 store locations. This is proof that Starlink hardware and software is functional at our store locations.

Graphic#1

(*Starlink Roof Mount - wall / eave mount - gen 2* 2023)

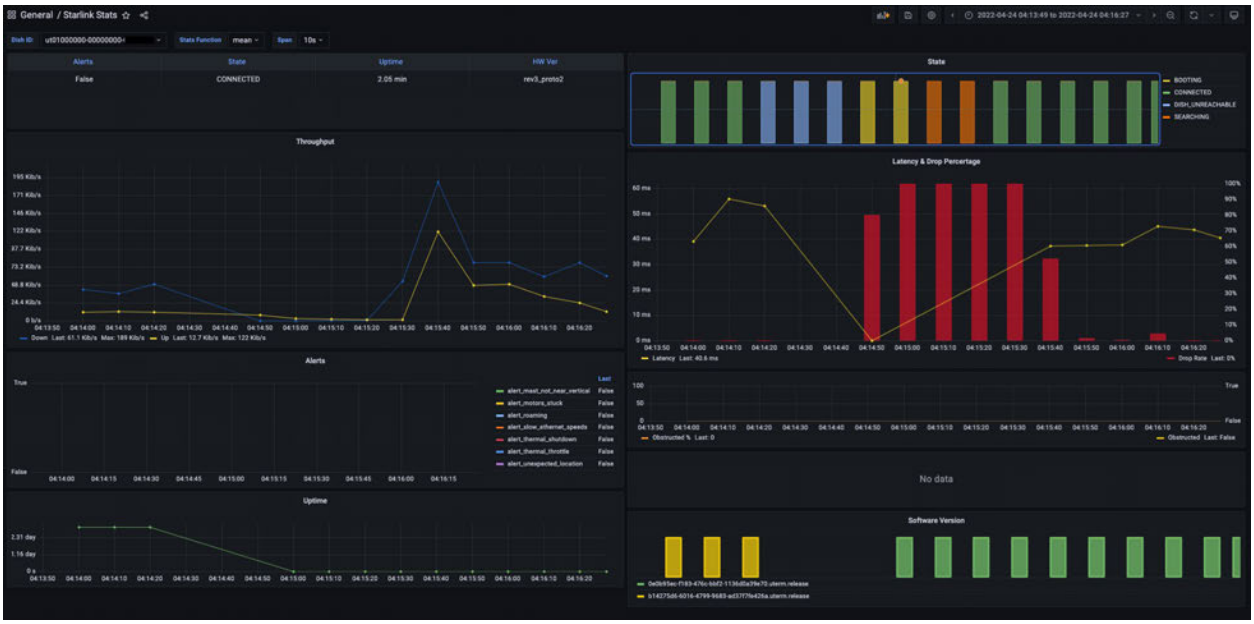
Graphic#2

(*Starlink Roof Mount - wall / eave mount - gen 2* 2023)

Graphic#3

(*Starlink setup and Installation Guide* 2023)

Graphic#4



(Rzzldzzl, 2023)

Risk Register

WESTERN GOVERNORS UNIVERSITY

Master of Science in IT Management (MSITM) Capstone Risk Register

SSM3: Design and Development

Asset	Threat/Vulnerability	Existing Controls	Likelihood	Consequence	Level of Risk	Risk Priority
Out of the box failure	Delay on installations	Additional Hardware Kits	Possible	Moderate	Medium	3
Harsh Weather Conditions	Delay on installation	Float Time In Schedule	Possible	Minor	Medium	3
Natural Disaster	Excessive Delay For Deployment	Float Time In Schedule	Rare	Major	High	1
Stakeholders Become Disengaged	Project Cancelation	Communication Plan	Rare	Major	Extreme	5
Individual Component Failure	Project Expense Rises	Contengency Reserve	Possible	Minor	Low	4
Project Exceeds Deadline	Project Cancellation	Project Management Plan	Rare	Major	High	5
Starliink Goes Out Of Business	Project Cancellation	SEC Filings Review	Rare	Major	Low	5
Starlink Raises Prices	Project Cancellation	Operational Efficiency	Almost Certa	Moderate	Medium	2
Project Team Knowledge	Delay on installations	Training Class	Rare	Minor	Low	4
Starlink Equipment Shortage	Delay on installations	Communication Plan	Rare	Moderate	Low	2
Concurrent Project Is Launched	Project Cancellation	Communication Plan	Rare	Minor	Low	3
Project Team Member Quits	Delay on installations	Two Additional Hires	Possible	Minor	Low	3
Project Exceeds Budget	Project Cancellation	Bottom-Up Analysis	Rare	Major	High	5
Project Team Member Injury	Resource Re-Allocation	Training Class	Possible	Minor	Low	3

Risk Register Explanation - I sent the risk register for the Starlink project between July 20, 2024, and July 31, 2024, which outlines all the initial risks I had identified during the project planning phase. The purpose of sending this document is to reinforce the diligence I had put into the project planning.

Qualitative Analysis

QUALITATIVE RISK ANALYSIS

		LIKELIHOOD		
		Almost Certain	Possible	Rare
Consequence	Major	0	0	5
	Moderate	1	1	1
	Minor	0	4	2
Red	These risks require a detailed action plan prior to occurrence.			
Yellow	These risks require monitoring throughout the project.			
Green	These risks will be addressed if they occur.			

Qualitative Analysis Explanation - I sent the Qualitative Analysis chart between July 20, 2024, and July 31, 2024, which displays the number of risks in the risk register that could have a high impact. In case any of these risks occurred, I needed an action plan in place beforehand. The chart also shows the number of issues in the register that were less critical and could be monitored without an immediate action plan.

Critical Risks W/ Plan

Critical Risks W/ Plan

Item	Action Plan
Harsh Weather Conditions	Proactively screen for bad weather when assigning the locations to which technicians will install that week.
Natural Disaster	Purchase an additional 100 Starlink Hardware Kits than the minimum needed for the project.
Stakeholders Become Disengaged	Use the scrum methodology that allows stakeholders to have a heavy hand in what is prioritized by the project team.
Individual Component Failure	Purchase spare components and have them pass our quality tests before storing them for needed use.
Project Exceeds Deadline	Use the scrum methodology to let us know how much progress has been made after each 30-day sprint and better judge if we need to catch up or ahead of schedule.
Starliink Goes Out Of Business	Review the SpaceX SEC Filings to see how well the business has been performing quarterly, semi-annually, and annually to get a better understanding of its true business state.
Project Team Member Quits	Hire two additional team members that will allow for reasonable time for replacement if someone abruptly quits.
Project Exceeds Budget	We will have a contingency reserve for unexpected expenses.
Project Team Member Injury	Provide adequate training, including emergency, safety, and installation training.

Critical Risks W/ Plan Explanation - I sent a chart to highlight the risks that were identified in the qualitative risk analysis and required an immediate action plan. Along with the chart, I shared a plan that I had developed to mitigate the risks beforehand or the steps that would be taken if the issue arose.

Cost Benefit Analysis

Cost Benefit Analysis & Mitigation

ID	Item	Est. Cost of Risk	Impact	Advanced Mitigation	Benefit of Advanced Mitigation
1	Out of the box failure	0,00 US$	Delays to the progress of the project.	Purchase an additional 100 Starlink Hardware Kits than the minimum needed for the project.	No delays the to progress of the project due to wait time to receive replacement equipment.
2	Harsh Weather Conditions	0,00 US$	Delays to the progress of the project.	Proactively screen for bad weather when assigning the locations to which technicians will install that week.	No delays to the project's progress due to delays caused by pausing installs due to weather conditions.
3	Natural Disaster	2500,00 US$	Delays to the progress of the project.	Purchase an additional 100 Starlink Hardware Kits than the minimum needed for the project.	The installation can occur as soon as feasible instead of waiting for whatever time Starlink has for equipment during our time of need.
4	Stakeholders Become Disengaged	Up to Entire Budget ($)	Project Cancellation	Use the scrum methodology that allows stakeholders to have a heavy hand in what is prioritized by the project team.	The project has the highest likelihood of completion, and the company has the best opportunity to get a return on its investment.
5	Individual Component Failure	Up to $2500	Delays to the progress of the project.	Purchase spare components and have them pass our quality tests before storing them for needed use.	No delays to the project's progress due to delays caused by individual components failing.
6	Project Exceeds Deadline	Minimum of $1,500,000	Project Cancellation	Use the scrum methodology to let us know how much progress has been made after each 30-day sprint and better judge if we need to catch up or ahead of schedule.	Increased quality output of the project team and an increase in stakeholder engagement.
7	Starlink Goes Out Of Business	Up to Entire Budget ($)	Project Cancellation	Review the SpaceX SEC Filings to see how well the business has been performing quarterly, semi-annually, and annually to get a better understanding of its true business state.	The project has the highest likelihood of completion, and the company has the best opportunity to get a return on its investment.
8	Starlink Price Increase	Varies	Could Exceed The Budget	Perform an analysis to ensure the stores with less traffic are not on the premium starlink business plan.	The impact of a price change is less felt.
9	Project Team Knowledge	0,00 US$	Delays to the progress of the project.	Provide adequate training, including emergency training, safety training, and installation training.	No delays to the project's progress due to the learning curve that comes with learning a new process and product.
10	Starlink Equipment Shortage	0,00 US$	Delays to the progress of the project.	Communicate with Starlink formally and ahead of time to ensure a proper understanding of what we are trying to accomplish along with what is feasible for SpaceX.	There were no delays to the project's progress due to Starlink not getting us the needed equipment in time.
11	Concurrent Project Is Launched	Up to Entire Budget ($)	Project Cancellation	Hire a Junior Project Manager that can take over in my place if the need arises.	The project can persist even if I am taken away from the project to persist with a new one.
12	Project Team Member Quits	Up to $100,000	Delays to the progress of the project.	Hire two additional team members that will allow for reasonable time for replacement if someone abruptly quits.	The project does not face any detriment to the project schedule.
13	Project Exceeds Budget	Up to Entire Budget ($)	Project Cancellation	We can have a contingency reserve for unexpected expenses.	Stakeholders are not troubled or pestered by constant requests for more money.
14	Project Team Member Injury	Varies	Delays to the progress of the project.	Provide adequate training, including emergency, safety, and installation training.	We do not face lawsuits, and the project can advance without detriment.

Cost Benefit Analysis Explanation - I sent the cost-benefit analysis chart to show the advanced mitigation measures I proposed to handle the identified risks ahead of time and to explain what benefits the company would potentially experience with the advanced mitigation.

Stakeholders Communication Plan

Stakeholders Communications Plan Example

Key Stakeholders	Role	Frequency	Communication Method	Notes	[redacted]
[redacted]	Chief Executive Officer	Weekly	Email/Video Conf.	Email for less critical updates, major critical updates require immediate video conference.	[redacted]
	Chief Informatiot Officer	Weekly	In-Person/Video Conf.	We will engage with the CIO weekly for all updates.	
	Chief Officer of Public Relations	Major Milestones	Email/Phone	We will engage Dana Koruvski regarding updates to the public about progress.	
	Corporate Operations Manager	Daily	Email/Phone/In-Person	We will speak with Olivia Banks daily, as she is the overseeing person for all operations in the company.	
	Chief Security Officer	Major Milestones	Email/Video Conf.	We will speak with Joanna Dark periodially to involve her to ensure our project has not breached any security policies.	

Stakeholders Communication Plan Explanation – I sent a visual that explains how communication would be facilitated throughout the project. This graphic displays the name of the stakeholder, the appropriate contacting time, the communication method that should be used, the frequency of communication, and their contact details. The purpose of sending this document is to reinforce the diligence I had put into the project planning.

Quantitative Risk Analysis

Quanitative Risk Analysis

Asset	Threat/Vulnerability	Existing Controls	Likelihood	Consequence	Level of Risk	Risk Priority	Cost Impact	Risk Responses
Stakeholders Become Disengaged	Project Cancelation	Communication Plan	Rare	Major	Extreme	5	Up to Entire Budget ($)	Hold an emergency meeting to regain Stakeholder Engagement
Individual Component Failure	Project Expense Rises	Contengency Reserve	Possible	Minor	Low	4	Up to $2500	Work with Starlink to find a root cause to create a permanent fix if the component failure trends.
Project Exceeds Deadline	Project Cancellation	Project Management Plan	Rare	Major	High	5	Up to Entire Budget ($)	Communicate with key stakeholders with evidence of the success thus far and remind them of the ROI.
Starliink Goes Out Of Business	Project Cancellation	SEC Filings Review	Rare	Major	Low	5	Minimum of $1,500,000	Cancel Service Immediately to save on any unnecessary expense if this happens.

Quantitative Risk Analysis Explanation – I sent a visual that shows how financially impactful the risks that were found to be critical could be to the project. The purpose of sending this document is to reinforce the diligence I had put into the project planning.

Assumptions Log

Assumption Log

ID	Category	Responsiblity	Due Date	Status	Actions	Assumptions
1	Human Resources	IT/Project Manager	03/15/2024	Open	Craft job requisition posts, and get them to HR to advertise.	We will be able to hire the necessary staff for the project.
2	Production	IT/Project Manager	03/01/2024	Open	Formally reach out to Starlink to get confirmation.	Starlink will allow for an order of 600 Starlink Kits at one time.
3	Finance	IT/Project Manager	1/31/2024	Open	Send project proposal for approval or denial inclusive of proposed budget.	The company has the money to take on this project.
4	Human Resources	Junior Project Man.	1/31/2024	Open	Identify the tenure of each of the key stakeholders.	Stakeholders will stay with the company throughout the project.
5	Reputation	Junior Project Man.	6/1/2024	Open	Create streams for feedback to be viable from our customers.	Customers will be satisfied.

Assumptions Log Explanation – I sent a visual that shows the assumptions that were made in the project. The purpose of sending this document is to reinforce the diligence I had put into the project planning.

Project Schedule/Timeline

ID	Task	Phase/Process	Duration	Start	Finish
1	Starlink Project Official Launch	Project Charter Completion - Project Launch	1 day	4/1/2024 8:00	4/1/2024 17:00
2	Call Starlink To Discuss Order	Project Management Execute Process Group	1 day	4/2/2024 8:00	4/2/2024 17:00
3	Starlink Equipment Gets Ordered	Project Management Execute Process Group	1 day	4/3/2024 8:00	4/3/2024 17:00
4	Starlink Equipment Arrives	Project Management Execute Process Group	6 days	4/15/2024 8:00	4/20/2024 17:00
5	Hire Project Team	Project Management Execute Process Group	12 days	4/1/2024 8:00	4/12/2024 17:00
6	Establish 24 Processes In PM Planning Process Group	Project Planning & Design Completion	29 days	4/2/2024 8:00	4/30/2024 17:00
7	Project Team Members Are Trained	Project Management Execute Process Group	16 days	4/15/2024 8:00	4/30/2024 17:00
8	Scrum Planning Meeting	Scrum Planning Meeting Meeting	1 day	5/1/2024 8:00	5/1/2024 17:00
9	Discuss Roadmap Of Sprint W/ Team	Scrum Initiation Phase	1 day	5/1/2024 8:00	5/1/2024 17:00
10	Plan The Current Sprint Details	Scrum Planning Phase	1 day	5/1/2024 8:00	5/1/2024 17:00
11	Installation Audits	Quality Review 1	10 days	5/15/2024 8:00	5/24/2024 17:00
12	Installations Begin	Scrum Execution Phase	30 days	5/1/2024 8:00	5/30/2024 17:00
13	Scrum Sprint Review Meeting	Scrum Review Phase	1 day	5/31/2024 8:00	5/31/2024 17:00
14	200 Installs Complete Milestone	MileStone - 200 Installations Complete	31 days	5/31/2024 8:00	6/30/2024 17:00
15	Scrum Planning Meeting	Scrum Planning Phase	1 day	6/1/2024 8:00	6/1/2024 17:00
16	Installation Audits	Quality Review 2	10 days	6/15/2024 8:00	6/24/2024 17:00
17	Installations Persist	Scrum Execution Phase	30 days	6/1/2024 8:00	6/30/2024 17:00
18	Scrum Sprint Review Meeting	Scrum Review 2	2 days	6/30/2024 8:00	7/1/2024 17:00
19	400 Installs Complete Milestone	MileStone - 400 Installations Complete	30 days	6/1/2024 8:00	6/30/2024 17:00
20	Plan The Current Sprint Details	Scrum Planning Phase	1 day	7/1/2024 8:00	7/1/2024 17:00
21	Quality Assessment	Quality Review 3	15 days	7/1/2024 8:00	7/15/2024 17:00
22	Installations Persist	Scrum Execution Phase	15 days	7/1/2024 8:00	7/15/2024 17:00
23	Scrum Sprint Review Meeting	Scrum Review 3	1 day	12/9/2023 8:00	12/9/2023 17:00
24	500 Installs Complete Milestone	MileStone - 500 Installations Complete	15 days	6/1/2024 8:00	6/15/2024 17:00
25	Scope Validation	Project Deliverable Acceptance	5 days	7/16/2024 8:00	7/20/2024 17:00
26	Room For Rework	Rework	44 days	7/20/2024 8:00	9/1/2024 17:00
27	Closure Tasks	Project Closure	30 days	9/1/2024 8:00	9/30/2024 17:00

Project Schedule/Timeline Explanation – I sent a visual showing the breakdown of each phase that would take place during the project, so that you had no confusion as to what would take place from the start of the project to the proposed end date of the project. The purpose of sending this document is to reinforce the diligence I had put into the project planning.

Project Plan

PROJECT PLAN

PROJECT NAME	Starlink Project	IT PROJECT MANAGER	Roscoe Govan
PROJECT SCOPE	The Starlink project's key objective is to successfully deploy the hardware and configure the software for Starlink service to the existing 500 GoodGrocery locations to serve as a network failover for each GoodGrocery store. Each location can expect the following as part of the deployments: 1 Starlink High-Performance Satellite Dish, 1 Starlink Power Supply, 1 cable that delivers power from the power supply to the Starlink satellite dish, and lastly, 1 proprietary cable that traverses from the power supply to our network switch for proper routing of data. The deliverables for this project will be a completed project, 500 locations with a functioning network failover solution, project documents, lessons learned, assumptions log, pert chart, and Gant chart. This project will not include anything not mentioned here in this scope statement.		
COMPANY	GoodGrocery		
CONTACT NAME	Roscoe Govan		
EMAIL	rgovan3@wgu.edu	PHONE	210-737-7047
START DATE	04/01/2024	END DATE	09/31/2024

SL. No.	ACTIVITY/ TASK NAME	Resources Assigned	START DATE	END DATE	DURATION (in days)	TASK STATUS
1	Starlink Project Official Launch	Project Proposal	Mon 4/1/24 8:00 AM	Mon 4/1/24 5:00 PM	1 day	Incomplete
2	Call Starlink to Discuss Order	Cellular Phone	Tue 4/2/24 8:00 AM	Tue 4/2/24 5:00 PM	1 day	Pending Project Start
3	Starlink Equipment Gets Ordered	Phone & Computer	Wed 4/3/24 8:00 AM	Wed 4/3/24 5:00 PM	1 day	Pending Project Start
4	Starlink Equipment Arrives	Open Room in Company Building For Storage	Mon 4/15/24 8:00 AM	Sat 4/20/24 5:00 PM	6 days	Pending Project Start
5	Hire Project Team	IT Manager & HR	Mon 4/1/24 8:00 AM	Fri 4/12/24 5:00 PM	12 days	Pending Project Start
6	Establish 24 Processes in PM Planning Process Group	IT Manager, Project Team, Stakeholders, Computer, Zoom	Tue 4/2/24 8:00 AM	Tue 4/30/24 5:00 PM	29 days	Pending Project Start
7	Project Team Members Are Trained	Starlink Training Videos, IT Manager, Instruction Manual, Company Policy Book	Mon 4/15/24 8:00 AM	Tue 4/30/24 5:00 PM	16 days	Pending Project Start
8	Scrum Planning Meeting	Computer, Zoom, Office Area	Wed 5/1/24 8:00 AM	Wed 5/1/24 5:00 PM	1 day	Pending Project Start
9	Discuss Roadmap of Sprint W/ Team	Computer, Zoom, Office Area	Wed 5/1/24 8:00 AM	Wed 5/1/24 5:00 PM	1 day	Pending Project Start
10	Plan The Current Sprint Details	Computer, Zoom, Office Area	Wed 5/1/24 8:00 AM	Wed 5/1/24 5:00 PM	1 day	Pending Project Start
11	Installation Audits	Computer, Zoom, Cell Phone	Wed 5/15/24 8:00 AM	Fri 5/24/24 5:00 PM	10 days	Pending Project Start
12	Installations Begin	IT Manager, Starlink Equipment, Network Engineers, Network Administrators, Junior Project Manager	Wed 5/1/24 8:00 AM	Thu 5/30/24 5:00 PM	30 days	Pending Project Start
13	Scrum Sprint Review Meeting	Computer, Zoom, Office Area	Fri 5/31/24 8:00 AM	Fri 5/31/24 5:00 PM	1 day	Pending Project Start
14	200 Installs Complete	IT Manager, Starlink Equipment, Network Engineers, Network Administrators, Junior Project Manager	Fri 5/31/24 8:00 AM	Sun 6/30/24 5:00 PM	31 days	Pending Project Start
15	Scrum Planning Meeting	Computer, Zoom, Office Area	Sat 6/1/24 8:00 AM	Sat 6/1/24 5:00 PM	1 day	Pending Project Start
16	Installation Audits	Computer, Zoom, Cell Phone	Sat 6/15/24 8:00 AM	Mon 6/24/24 5:00 PM	10 days	Pending Project Start
17	Installations Persist	IT Manager, Starlink Equipment, Network Engineers, Network Administrators, Junior Project Manager	Sat 6/1/24 8:00 AM	Sun 6/30/24 5:00 PM	30 days	Pending Project Start
18	Scrum Sprint Review Meeting	Computer, Zoom, Office Area	Sun 6/30/24 8:00 AM	Mon 7/1/24 5:00 PM	2 days	Pending Project Start
19	400 Installs Complete	IT Manager, Starlink Equipment, Network Engineers, Network Administrators, Junior Project Manager	Sat 6/1/24 8:00 AM	Sun 6/30/24 5:00 PM	30 days	Pending Project Start
20	Plan The Current Sprint Details	Computer, Zoom, Office Area	Mon 7/1/24 8:00 AM	Mon 7/1/24 5:00 PM	1 day	Pending Project Start
21	Quality Assessment	Computer, Zoom, Cell Phone	Mon 7/1/24 8:00 AM	Mon 7/15/24 5:00 PM	15 days	Pending Project Start
22	Installations Persist	IT Manager, Starlink Equipment, Network Engineers, Network Administrators, Junior Project Manager	Mon 7/1/24 8:00 AM	Mon 7/15/24 5:00 PM	15 days	Pending Project Start
23	Scrum Review 3	Computer, Zoom, Office Area	Wed 7/10/24 8:00 AM	Mon 7/15/24 5:00 PM	5 day	Pending Project Start
24	Milestone - 500 Installations Complete	IT Manager, Starlink Equipment, Network Engineers, Network Administrators, Junior Project Manager	Wed 7/10/24 8:00 AM	Mon 7/15/24 5:00 PM	15 days	Pending Project Start
25	Project Deliverable Acceptance	Computer, Zoom, Office Area, Checklist	Mon 7/1/24 8:00 AM	Mon 7/15/24 5:00 PM	5 days	Pending Project Start
Rework	Room For Rework	Computer, Zoom, Office Area	Sat 7/20/24 8:00 AM	Sun 9/1/24 5:00 PM	44 days	Pending Project Start
Project Closure	Closure Tasks	Computer, Zoom, Office Area	Sun 9/1/24 8:00 AM	Mon 9/30/24 5:00 PM	30 days	Pending Project Start

(Project plan template for word & excel (free download): Polaris 2023)

Project Plan Explanation – I sent a document called the "Project Plan," which is a document that outlines much of what we have already discussed but centralizes all the information. This document includes the Communication Plan, Project Scope Statement, Project Terms for Closure, Assumption Log, Project Phases, Project Schedule/Timeline, Dependencies, Risk Factors, Milestones, and Documentation Deliverables. the project plan is a critical document in the project as it is the document that you can go to for nearly everything entailed in the project.

Formative Evaluation Results and Revisions

We relied on the project timeline and project milestones to track the project's progress, and these milestones were assigned a percentage of completion. The key milestones that we tracked were:

1. Project Charter Completed - Project 12.5% Complete
2. Starlink Equipment Received - Project 25% Complete
3. Project Team Trained - Project 37.5% Complete
4. 200 Installations Complete - Project 50% Complete
5. 400 Installations Complete - Project 62.5% Complete
6. 500 Installations Complete - Project 75% Complete
7. Project Deliverables Accepted - Project 87.5% Complete
8. Project Closure Tasks Complete – Project 100% Complete

There were no project revisions nor changes that were done based on the quality enforcements since all efforts went without flaw. To ensure the successful completion of installations, I used checklists and audits. The checklists were designed to ensure that all required tasks were completed. Both network architects and telecommunications technicians were responsible for completing their respective checklists, which were then emailed to me for review. The telecommunications technician's checklist had different items than the network architect's, but the architect's checklist included all the technician's items. To provide better clarity, I have included a sample below.

Telecommunications Technician Checklist

Enter Store Number Here:

Date	Task Completed	Issues Encountered	Reported Issue	Issue Explanation	Task Complete/Not Complete
##/##/####	Installed Starlink Hardware Dish?	Y/N	Y/N		
##/##/####	Confirmed Starlink Hardware Dish is pointed towards space?	Y/N	Y/N		
##/##/####	Installed power cord that goes from Starlink Dish to Power Supply?	Y/N	Y/N		
##/##/####	Confirmed no damage to the power cord that goes from Starlink Dish to Power Supply?	Y/N	Y/N		
##/##/####	Installed Starlink Power Supply in a well ventilated area?	Y/N	Y/N		
##/##/####	Confirmed Starlink Power Supply indicator light showed green?	Y/N	Y/N		
##/##/####	Installed the Starlink Power Supply power cord to working outlet?	Y/N	Y/N		
##/##/####	Confirmed Starlink Proprietary Cable was functional and connected to network switch?	Y/N	Y/N		
##/##/####	Worked with Network Architect until Starlink was confirmed online?	Y/N	Y/N		
##/##/####	Sent clear photos of each Starlink component after install?	Y/N	Y/N		

Network Architect Checklist

Enter Store Number Here:

Date	Task Completed	Issues Encountered	Reported Issue	Issue Explanation	Task Complete/Not Complete
##/##/####	Validated Starlink Hardware Dish?	Y/N	Y/N		
##/##/####	Confirmed Starlink Hardware Dish is pointed towards space?	Y/N	Y/N		
##/##/####	Validated power cord that goes from Starlink Dish to Power Supply?	Y/N	Y/N		
##/##/####	Confirmed no damage to the power cord that goes from Starlink Dish to Power Supply?	Y/N	Y/N		
##/##/####	Validated Starlink Power Supply in a well ventilated area?	Y/N	Y/N		
##/##/####	Confirmed Starlink Power Supply indicator light showed green?	Y/N	Y/N		
##/##/####	Validated the Starlink Power Supply power cord to working outlet?	Y/N	Y/N		
##/##/####	Confirmed Starlink Proprietary Cable was functional and connected to network switch?	Y/N	Y/N		
##/##/####	Worked with Telecommunication Technician until Starlink was confirmed online?	Y/N	Y/N		
##/##/####	Archived clear photos of each Starlink component after install?	Y/N	Y/N		
##/##/####	Configured the Starlink equipment to work as a failover for the store?	Y/N	Y/N		
##/##/####	Compared configuration to a known working site?	Y/N	Y/N		
##/##/####	Updated knowledge base with anything new?	Y/N	Y/N		
##/##/####	Educated & Involved network administrators?	Y/N	Y/N		
##/##/####	Checked Starlink Portal for SN accuracy?	Y/N	Y/N		
##/##/####	Checked Starlink Portal to ensure no obstruction present?	Y/N	Y/N		

Summative Evaluation Plan and Results

Throughout the Starlink project, we used several methods to evaluate the project's completion and deliverables. These methods included checklists, check sheets, and inspections. We had multiple lists to assess the quality of Starlink equipment installations, and each technical worker had a checklist. I conducted in-person inspections to verify the submitted checklists. Additionally, I manually checked the equipment's expected functionality weekly for the stores marked complete on our list. By having these QA measures in place, we attained the successful signed off on all deliverables an entire month ahead of schedule. The Starlink project was a tremendous success and can now serve as an example of proper planning leading to exceptional results for future projects.

Stakeholder Communication Plan and Reports

Stakeholders Communications Plan Example

Key Stakeholders	Role	Frequency	Communication Method	Notes	Phone Number	Email
[redacted]	Chief Executive Officer	Weekly	Email/Video Conf.	Email for less critical updates, major critical updates require immediate video conference.	[redacted]	[redacted]
	Chief Information Officer	Weekly	In-Person/Video Conf.	We will engage with the CIO weekly for all updates.		
	Chief Officer of Public Relations	Major Milestones	Email/Phone	We will engage Dana Korverski regarding updates to the public about progress.		
	Corporate Operations Manager	Daily	Email/Phone/In-Person	We will speak with Olivia Banks daily, as she is the overseeing person for all operations in the company.		
	Chief Security Officer	Major Milestones	Email/Video Conf.	We will speak with Joanna Dark periodially to involve her to ensure our project has not breached any security policies		

The key stakeholders in the Starlink project were [redacted] (CEO), [redacted] (CIO), [redacted] (Chief Officer of Public Relations), [redacted] (Corporate Operations Manager), and [redacted] (Chief Security Officer). I used the Mendolow matrix to help identify power and influence among stakeholders, and I set a default frequency to communicate with each of you as key stakeholders based on my determination, with some exceptions.

Based on the crafted communications plan I communicated with Mr. [redacted] every week from the start of the project. When I reached out to Mr. [redacted] via video conference to provide him with both highly critical and less critical details that had come about

throughout the week. At the end of every meeting, he received an email with keynotes that I had discussed with him via video conference for reference purposes.

Based on the crafted communications plan I communicated with Mr. [redacted] on a weekly basis as well from the start of the Starlink project. Adam was often part of the same meetings I had with [redacted] and would receive the same information that Mr. [redacted] did. The difference in the conversational talking point in the meetings between Mr. [redacted] and [redacted] were that I spoke with both about aspects that could impact business, and project progression, but mainly Mr. [redacted] when my explanations involved technical details such as network statistics as he is the CIO of the company and was much more suited for a technical discussion where as Mr. [redacted] primal responsibility is the overall company detriment or benefit.

Based on the crafted communications plan I communicated with [redacted] at every milestone in the project that could be something we wanted to make public to our business partners, customers, or both. Ms. [redacted] is the Chief Officer of Public Relations, so she was best fitted for this assessment, and the decision to publish our progress to business partners, customers, or both, was ultimately an action taken after she ensured the awareness of Mr. [redacted] I have documented the different milestones that were achieved during my conversations with Ms. [redacted] These are as follows:

- Milestone 1: Project Charter Completion (Completed April 1, 2024) - I sent an email to Ms. [redacted] to let her know that the project was official. I did this because she would be responsible for creating public advertisements announcing that our company has adopted Starlink.

- Milestone 7: 200 Installations Complete (Completed May 31, 2024) - On May 31, 2024, I spoke with Ms. [redacted] over the phone and via email to inform her that we had completed 200 installations for the Starlink project. I also provided her with the percentage of completion that the project had achieved. This information will help her decide whether to announce to external stakeholders that the Starlink project is (X%) complete to generate excitement.

- Milestone 8: 400 Installations Complete (Completed June 30, 2024) - On June 30, 2024, I spoke with Ms. [redacted] over the phone and via email to inform her that we had completed 400 installations for the Starlink project. I also provided her with the percentage of completion that the project had achieved. This information will help her decide whether to announce to external stakeholders that the Starlink project is (X%) complete to generate excitement.

- Milestone 9: 500 Installations Complete (Completed July 15, 2024) - On July 15, 2024, I spoke with Ms. [redacted] over the phone and via email to inform her that we had completed 500 installations for the Starlink project. I also provided her with the percentage of completion that the project had achieved. This information will help her decide whether to announce to external stakeholders that the Starlink project is (X%) complete to generate excitement.

- Milestone 11: Project Closure (Completed July 31, 2024) – On July 31, 2024, I formally advised Ms. [redacted] via phone and email that the project was closed and that the Starlink project was a success. This was done to enable her to communicate with all stakeholder groups in a public manner that is appropriate for our company's leadership.

Based on the crafted communications plan I communicated daily with [redacted] the Corporate Operations Manager. Ms. [redacted] and I communicated via email, phone, or in-person, depending on the level of criticality/sensitivity to the information needing to be shared. We spoke daily, so I did not outline the conversation that was held every single day here, however, I have documented all critical communications we had during the project. These are as follows:

- Milestone 1: Project Charter Completion (Completed April 1, 2024) - I sent an email to Ms. [redacted] to let her know that the project was official. I did this because her job entails knowing all aspects of the company that could potentially impact other areas of the business.
- Milestone 11 - Project Closure *(July 31-2024)* – I emailed Ms. [redacted] to advise that the project had concluded and that it was a success. I provided her a summary of what changes the new technology would bring, so she could be knowledgeable when explaining to anyone inquiring.

Based on the crafted communications plan, I communicated with Ms. [redacted] only at milestones like Ms. [redacted] however, our conversations contained different content. Ms. [redacted] oversees technological security for our company, so my messages to her were in regard to

checking whether or not anything we had done violated security within the company at each milestone for proper assessment and correction. This was done to ensure we did not unknowingly create security vulnerabilities for the company. I have documented the milestones achieved during my conversations with Ms. [redacted] These are as follows:

- Milestone 7: 200 Installations Complete (Completed May 31, 2024) - On May 31, 2024, I spoke with Ms. [redacted] via email to inform her that we had completed 200 installations for the Starlink project. This communication was more of an inquiry to request her expert input regarding whether any new vulnerabilities that seemed to be tied to our project were present that I needed to address.

- Milestone 8: 400 Installations Complete (Completed June 30, 2024) - On June 30, 2024, I spoke with Ms. [redacted] via email to inform her that we had completed 400 installations for the Starlink project. This communication was more of an inquiry to request her expert input regarding whether any new vulnerabilities that seemed to be tied to our project were present that I needed to address.

- Milestone 9: 500 Installations Complete (Completed July 15, 2024) - On July 15, 2024, I spoke with Ms. [redacted] via email to inform her that we had completed 400 installations for the Starlink project. This communication was more of an inquiry to request her expert input regarding whether any new vulnerabilities that seemed to be tied to our project were present that I needed to address.

- Milestone 11: Project Closure (Security Audit) (Completed July 24, 2024) – On July 24, 2024, I formally advised Ms. [redacted] via phone and email that the project would be closed in a few days. This allowed her to conduct a final security audit of our company regarding security before closure.

The Starlink project's Stakeholder communication plan proved very effective as you, as stakeholders, expressed consistent satisfaction throughout the entire project due to your level of engagement. Additional aspects could be added to the communications plan for future projects. Still, as I have come to find, projects can be filled with an unfathomable number of tasks to complete daily, weekly, or monthly. Therefore, the best idea is to work towards simplicity, making anything more straightforward to execute constantly. This communications plan was simple and easy to follow.

Ongoing Support and Maintenance

During the initial proposal, I knew we would need staff to support and maintain the deployment after completing the project. So, I accounted for that need when requesting human resources. Now that the project has concluded, as we had discussed in the proposal, we have two network architects remaining after the additional three network architects on contracts left after their contracts ended. Those two network architects will act in senior roles that advise the three network administrators that remained upon project conclusion. The network administrators will be handling configuration issues, updates, and more at any store with the new Starlink solution. The two network architects will assist in training the network administrators until they are

reasonably self-sufficient regarding troubleshooting the store networks. The network architects have added much to the knowledge base for the network administrators to use for their work, so the training should be swift. The telecommunications technicians should be converted to work for the company full-time to assist the network administrators with the upkeep of the store networks. The junior project manager will serve as the supervisor of the other technical professionals I just mentioned, unless assigned to a project. These decisions will lead to business maintaining the 99.999% availability we are now moving towards.

Resources for Post-Implementation Support

Our stores are not open 24/7, so maintenance can be done after-hours, from 11:00 PM – 6:00 AM in the respective time zone. To support our store networks, we will need an available support team which will require financial allocation. The remaining telecommunications technicians will cost us $280,000 annually, as I proposed a salary of $40,000 for each of them, and we have seven. The network administrators will cost us $$180,000 annually, as they were offered a salary of $60,000, and we have three of them. The network architects will cost us $160,000 annually, as I proposed a salary of $80,000, and we have two of them remaining upon project completion. The junior project manager will cost us $100,000, as I proposed their salary at that rate upon hire. As a reminder, though this individual is a project manager, he will serve as a supervisor of the other technical professionals unless assigned to a project. The support team I am proposing is critical to our business continuity and the possibility of seeing the return on investment for the Starlink project we completed. Though this may sound like a lot of money, please remember that the Starlink project was over a two-million-dollar investment, and the maintenance for upkeep is approximately $720,000. The $720,000 would not be considered an expense when we now save four million dollars annually; it is simply an operations expense.

Short- and Long-Term Maintenance Plan

Short-Term & Long-Term Maintenance Plan				
Date	Type of Maintenance	Responsible Party	Results	Feedback
1-Jan-25	Network Hardware Check	Telecommunications Technicians		
1-Jan-25	Network Software Check	Network Administrators		
1-Jan-25	Network Failover Test	Network Architects		
1-Jan-25	Availability Evaluation (%)	Network Architects		
1-Jan-25	Staff Knowledge Test	Junior Project Manager (Supervisor)		
1-Jan-25	Starlink Communication	Junior Project Manager (Supervisor)		
1-Jan-25	Starlink SEC Filings Assessment	Junior Project Manager (Supervisor)		
1-Mar-25	Network Hardware Check	Telecommunications Technicians		
1-Mar-25	Network Software Check	Network Administrators		
1-Mar-25	Network Failover Test	Network Architects		
1-Mar-25	Availability Evaluation (%)	Network Architects		
1-Mar-25	Staff Knowledge Test	Junior Project Manager (Supervisor)		
1-Mar-25	Starlink Communication	Junior Project Manager (Supervisor)		
1-Mar-25	Starlink SEC Filings Assessment	Junior Project Manager (Supervisor)		
1-Jun-25	Network Hardware Check	Telecommunications Technicians		
1-Jun-25	Network Software Check	Network Administrators		
1-Jun-25	Network Failover Test	Network Architects		
1-Jun-25	Availability Evaluation (%)	Network Architects		
1-Jun-25	Staff Knowledge Test	Junior Project Manager (Supervisor)		
1-Jun-25	Starlink Communication	Junior Project Manager (Supervisor)		
1-Jun-25	Starlink SEC Filings Assessment	Junior Project Manager (Supervisor)		
1-Aug-25	Network Hardware Check	Telecommunications Technicians		
1-Aug-25	Network Software Check	Network Administrators		
1-Aug-25	Network Failover Test	Network Architects		
1-Aug-25	Availability Evaluation (%)	Network Architects		
1-Aug-25	Staff Knowledge Test	Junior Project Manager (Supervisor)		
1-Aug-25	Starlink Communication	Junior Project Manager (Supervisor)		
1-Aug-25	Starlink SEC Filings Assessment	Junior Project Manager (Supervisor)		
1-Oct-25	Network Hardware Check	Telecommunications Technicians		
1-Oct-25	Network Software Check	Network Administrators		
1-Oct-25	Network Failover Test	Network Architects		
1-Oct-25	Availability Evaluation (%)	Network Architects		
1-Oct-25	Staff Knowledge Test	Junior Project Manager (Supervisor)		
1-Oct-25	Starlink Communication	Junior Project Manager (Supervisor)		
1-Oct-25	Starlink SEC Filings Assessment	Junior Project Manager (Supervisor)		
1-Dec-25	Network Hardware Check	Telecommunications Technicians		
1-Dec-25	Network Software Check	Network Administrators		
1-Dec-25	Network Failover Test	Network Architects		
1-Dec-25	Availability Evaluation (%)	Network Architects		
1-Dec-25	Staff Knowledge Test	Junior Project Manager (Supervisor)		
1-Dec-25	Starlink Communication	Junior Project Manager (Supervisor)		
1-Dec-25	Starlink SEC Filings Assessment	Junior Project Manager (Supervisor)		

Both short-term and long-term maintenance will be performed as explained in the section titled “Ongoing Support and Maintenance.” As far as a breakdown of annual maintenance I have provided a visual breakdown of how these paid IT professionals will be ensuring our technical business continuity. There will be seven tasks that will be performed every three months outside of the direct support that is provided daily by the retained IT professionals. The seven tasks that will be performed every three months are the following:

Network Hardware Check: The telecommunications technicians we are retaining will physically evaluate all hardware in the field to ensure that it is still intact and has not faced any damage or detriment for any reason.

Network Software Check: The network administrators will ensure that the software configuration regarding the primary internet connection and the Starlink equipment is up to date and implemented as expected.

Network Failover Test: The network architects will force the data in stores over to the Starlink solution to confirm the reliability of Starlink and report the statistics to me so I can re-enforce our business resiliency to you as the original stakeholders of the Starlink project.

Availability Evaluation: The junior project manager acting as a supervisor will assess our progress towards achieving the industry standard of 99.999% availability and advise what is holding us back from acting and eliminating the constraint.

Staff Knowledge Check: The junior project manager acting as a supervisor will distribute a test that includes all the knowledge that should have been attained regarding the hardware, software, and functionality of the Starlink solution up until this point. We will keep these archives to ensure we can track the increased competence of our staff.

Starlink Communication: The junior project manager acting as a supervisor will contact Starlink to learn about planned changes or new advancements regarding their technology that we must be aware of.

`Starlink SEC Filings Assessment: The junior project manager who will be acting as a supervisor will review any new SEC Filings posted regarding Starlink to identify if there has been a change in the company's positive trajectory.

Post-Implementation Project Summary

As of January 1, 2024, our company was losing money because our stores needed network redundancy. So, I researched different providers and found that Starlink was the best solution for our business because it is more affordable and provides better service quality than other providers. I presented my proposal to you as company leadership back on January 7, 2024. The proposal included all critical aspects an enterprise needs to consider before approving or

denying a proposed project. I provided you with a needs analysis, cost-benefit analysis, risk register, project plan, communications plan, and other critical documents to provide my level of diligence and my ability to be trusted with a financial investment towards something that would better our future. Upon receipt of the proposal, a few months went by and the project was officially approved and I received the project charter on the start date of the project which was April 1, 2024 and immediately began the project. The project ended up being a success and closed ahead of time after you all signed off on the project deliverables back on July 20, 2024. I am appreciative that you as leaders were able to see both my potential and the potential of the project itself, now everyone involved including our customers can enjoy the return on investment. Thank you for your collaboration.

Documentation Deliverables

The Starlink project's primary deliverables included five hundred stores with validated Starlink hardware kits installed and validated Starlink service. The Starlink project also had several documents that were required to be delivered at project closure, including the cost-benefit analysis, itemized cost breakdown, risk register, qualitative and quantitative risk analysis, timeline, scope statement, milestones, project launch details, implementation strategy, documentation and hardware/software deliverables, project phases, risk mitigation, risk factors, project plan, and the evaluation framework documents. I provided the stated documentation and two examples below to help identify some critical documents mentioned.

Document Example 1

PROJECT PLAN

PROJECT NAME	Starlink Project	IT PROJECT MANAGER	Roscoe Govan
PROJECT SCOPE	The Starlink project's key objective is to successfully deploy the hardware and configure the software for Starlink service to the existing 500 GoodGrocery locations to serve as a network failover for each GoodGrocery store. Each location can expect the following as part of the deployments: 1 Starlink High-Performance Satellite Dish, 1 Starlink Power Supply, 1 cable that delivers power from the power supply to the Starlink satellite dish, and lastly, 1 proprietary cable that traverses from the power supply to our network switch for proper routing of data. The deliverables for this project will be a completed project, 500 locations with a functioning network failover solution, project documents, lessons learned, assumptions log, pert chart, and Gant chart. This project will not include anything not mentioned here in this scope statement.		
COMPANY	GoodGrocery		
CONTACT NAME	Roscoe Govan		
EMAIL	rgovan3@wgu.edu	PHONE	210-737-7047
START DATE	04/01/2024	END DATE	09/31/2024

SL. No.	ACTIVITY/ TASK NAME	Resources Assigned	START DATE	END DATE	DURATION (in days)	TASK STATUS
1	Starlink Project Official Launch	Project Proposal	Mon 4/1/24 8:00 AM	Mon 4/1/24 5:00 PM	1 day	Incomplete
2	Call Starlink to Discuss Order	Cellular Phone	Tue 4/2/24 8:00 AM	Tue 4/2/24 5:00 PM	1 day	Pending Project Start
3	Starlink Equipment Gets Ordered	Phone & Computer	Wed 4/3/24 8:00 AM	Wed 4/3/24 5:00 PM	1 day	Pending Project Start
4	Starlink Equipment Arrives	Open Room In Company Building For Storage	Mon 4/15/24 8:00 AM	Sat 4/20/24 5:00 PM	6 days	Pending Project Start
5	Hire Project Team	IT Manager & HR	Mon 4/1/24 8:00 AM	Fri 4/12/24 5:00 PM	12 days	Pending Project Start
6	Establish 24 Processes in PM Planning Process Group	IT Manager, Project Team, Stakeholders, Computer, Zoom	Tue 4/2/24 8:00 AM	Tue 4/30/24 5:00 PM	29 days	Pending Project Start
7	Project Team Members Are Trained	Starlink Training Videos, IT Manager, Instruction Manual, Company Policy Book	Mon 4/15/24 8:00 AM	Tue 4/30/24 5:00 PM	16 days	Pending Project Start
8	Scrum Planning Meeting	Computer, Zoom, Office Area	Wed 5/1/24 8:00 AM	Wed 5/1/24 5:00 PM	1 day	Pending Project Start
9	Discuss Roadmap of Sprint W/ Team	Computer, Zoom, Office Area	Wed 5/1/24 8:00 AM	Wed 5/1/24 5:00 PM	1 day	Pending Project Start
10	Plan The Current Sprint Details	Computer, Zoom, Office Area	Wed 5/1/24 8:00 AM	Wed 5/1/24 5:00 PM	1 day	Pending Project Start
11	Installation Audits	Computer, Zoom, Cell Phone	Wed 5/15/24 8:00 AM	Fri 5/24/24 5:00 PM	10 days	Pending Project Start
12	Installations Begin	IT Manager, Starlink Equipment, Network Engineers, Network Administrators, Junior Project Manager	Wed 5/1/24 8:00 AM	Thu 5/30/24 5:00 PM	30 days	Pending Project Start
13	Scrum Sprint Review Meeting	Computer, Zoom, Office Area	Fri 5/31/24 8:00 AM	Fri 5/31/24 5:00 PM	1 day	Pending Project Start
14	200 Installs Complete	IT Manager, Starlink Equipment, Network Engineers, Network Administrators, Junior Project Manager	Fri 5/31/24 8:00 AM	Sun 6/30/24 5:00 PM	31 days	Pending Project Start
15	Scrum Planning Meeting	Computer, Zoom, Office Area	Sat 6/1/24 8:00 AM	Sat 6/1/24 5:00 PM	1 day	Pending Project Start
16	Installation Audits	Computer, Zoom, Cell Phone	Sat 6/15/24 8:00 AM	Mon 6/24/24 5:00 PM	10 days	Pending Project Start
17	Installations Persist	IT Manager, Starlink Equipment, Network Engineers, Network Administrators, Junior Project Manager	Sat 6/1/24 8:00 AM	Sun 6/30/24 5:00 PM	30 days	Pending Project Start
18	Scrum Sprint Review Meeting	Computer, Zoom, Office Area	Sun 6/30/24 8:00 AM	Mon 7/1/24 5:00 PM	2 days	Pending Project Start
19	400 Installs Complete	IT Manager, Starlink Equipment, Network Engineers, Network Administrators, Junior Project Manager	Sat 6/1/24 8:00 AM	Sun 6/30/24 5:00 PM	30 days	Pending Project Start
20	Plan The Current Sprint Details	Computer, Zoom, Office Area	Mon 7/1/24 8:00 AM	Mon 7/1/24 5:00 PM	1 day	Pending Project Start
21	Quality Assessment	Computer, Zoom, Cell Phone	Mon 7/1/24 8:00 AM	Mon 7/15/24 5:00 PM	15 days	Pending Project Start
22	Installations Persist	IT Manager, Starlink Equipment, Network Engineers, Network Administrators, Junior Project Manager	Mon 7/1/24 8:00 AM	Mon 7/15/24 5:00 PM	15 days	Pending Project Start
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24	Milestone - 500 Installations Complete	IT Manager, Starlink Equipment, Network Engineers, Network Administrators, Junior Project Manager	Wed 7/10/24 8:00 AM	Mon 7/15/24 5:00 PM	15 days	Pending Project Start
25	Project Deliverable Acceptance	Computer, Zoom, Office Area, Checklist	Mon 7/1/24 8:00 AM	Mon 7/15/24 5:00 PM	5 days	Pending Project Start
Rework	Room For Rework	Computer, Zoom, Office Area	Sat 7/20/24 8:00 AM	Sun 9/1/24 5:00 PM	44 days	Pending Project Start
Project Closure	Closure Tasks	Computer, Zoom, Office Area	Sun 9/1/24 8:00 AM	Mon 9/30/24 5:00 PM	30 days	Pending Project Start

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Document Example 2

WESTERN GOVERNORS UNIVERSITY

Master of Science in IT Management (MSITM) Capstone Risk Register

SSM3: Design and Development

Asset	Threat/Vulnerability	Existing Controls	Likelihood	Consequence	Level of Risk	Risk Priority
Out of the box failure	Delay on installations	Additional Hardware Kits	Possible	Moderate	Medium	3
Harsh Weather Conditions	Delay on installation	Float Time In Schedule	Possible	Minor	Medium	3
Natural Disaster	Excessive Delay For Deployment	Float Time In Schedule	Rare	Major	High	1
Stakeholders Become Disengaged	Project Cancelation	Communication Plan	Rare	Major	Extreme	5
Individual Component Failure	Project Expense Rises	Contengency Reserve	Possible	Minor	Low	4
Project Exceeds Deadline	Project Cancellation	Project Management Plan	Rare	Major	High	5
Starliink Goes Out Of Business	Project Cancellation	SEC Filings Review	Rare	Major	Low	5
Starlink Raises Prices	Project Cancellation	Operational Efficiency	Almost Certa	Moderate	Medium	2
Project Team Knowledge	Delay on installations	Training Class	Rare	Minor	Low	4
Starlink Equipment Shortage	Delay on installations	Communication Plan	Rare	Moderate	Low	2
Concurrent Project Is Launched	Project Cancellation	Communication Plan	Rare	Minor	Low	3
Project Team Member Quits	Delay on installations	Two Additional Hires	Possible	Minor	Low	3
Project Exceeds Budget	Project Cancellation	Bottom-Up Analysis	Rare	Major	High	5
Project Team Member Injury	Resource Re-Allocation	Training Class	Possible	Minor	Low	3

- Graphic #1 is a visual display of one Starlink deployment at store number eighty-five.

- Graphic #2 is a visual display of the nuts and bolts that came with the Starlink hardware kits the project team installed.

- Graphic #3 is a visual display of the Starlink Dish and a first look at the power supply our hardware kits came with. We installed these power supplies, which power the Starlink satellite dish, at every store.

- Graphic #4 is a visual of our new Starlink software as a dashboard that allows us to track how well Starlink operates when the primary fiber connection goes down for any reason. This software is centralized software that will enable us to monitor the service quality of Starlink flowing through all 500 store locations. This is proof that Starlink hardware and software is functional at our store locations.

Graphic#1

(*Starlink Roof Mount - wall / eave mount - gen 2* 2023)

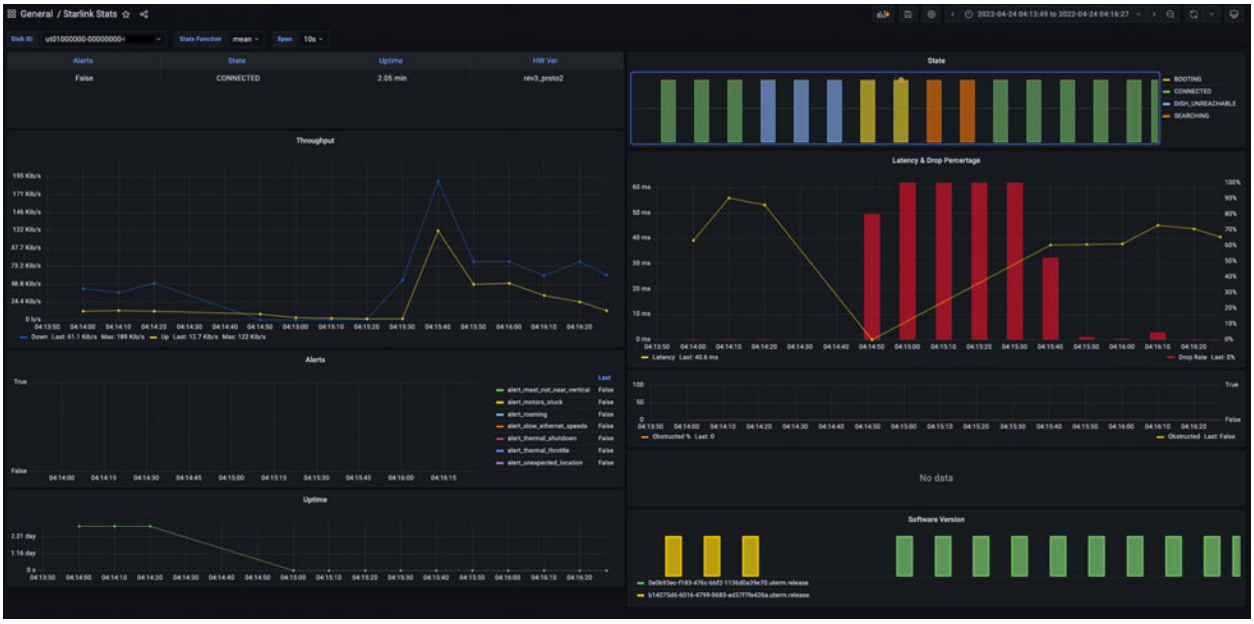
Graphic#2

(*Starlink Roof Mount - wall / eave mount - gen 2* 2023)

Graphic#3

(*Starlink setup and Installation Guide* 2023)

Graphic#4



(Rzzldzzl, 2023)

Success Criteria for Each Project Outcome

- **Project Charter Success Criteria:** To meet these criteria, we needed to receive the physical project charter documents with validated signatures from you all as the key stakeholders. The document you all provided me with allowed me to hold sufficient authority within our company to formally begin the project. On April 1, 2024, I received the document authorizing me to act as the primary project manager for the Starlink project. I officially initiated the project and made it a recognized effort within the organization.
- **Starlink Equipment Received Success Criteria** To meet these criteria, we needed to make sure that all the necessary components were present in every Starlink hardware kit. I

can confirm that on April 17, 2024, we received all the Starlink equipment that I had ordered on April 1, 2024, and that all the expected components were included.

- **Project Team Trained Success Criteria:** To meet these criteria, we needed to ensure that every member of the project team was able to complete 14 days of training without missing a single day. On April 31, 2024, I confirmed that all 15 newly hired members of the project team had completed their 14-day training program, which they started on April 15, 2024.
- **200 Installations Success Criteria:** To meet these criteria, we needed to confirm that the hardware and software were properly installed and functioning as expected at each of the 200 store locations. On May 31, 2024, I was able to confirm that both the hardware and software involved in the first sprint had been successfully installed and were working as expected at all 200 store locations.
- **400 Installations Success Criteria:** To meet these criteria, we needed to ensure that both the hardware and software were installed and functioning correctly at each store location. On June 31, 2024, during the second sprint, I validated that the hardware and software had been successfully installed and were functioning correctly at all 200 store locations.
- **500 Installations Success Criteria:** To meet these criteria, we needed to ensure that the Starlink hardware and software had been installed and were functioning properly at each of the 100 store locations. On July 15, 2024, I was able to validate that the hardware and software involved in the first sprint had been completed as expected at all 100 store locations.
- **Project Deliverables Accepted Success Criteria:** For this to have been considered a success, we must have obtained the signatures of all critical stakeholders with formal acknowledgment in writing that they have read and understand the documents at hand. As

of July 16, 2024, through July 20, 2024, I involved you as stakeholders for final scope validation, where you could analyze the deliverables and sign off on each. On July 20, 2024, the document showing your understanding of the documents and the scope validation documents were successfully signed, formally acknowledging the acceptance of all deliverables.

- **Project Closure Tasks Complete Success Criteria:** For this to have been considered a success, the administrative documents and project documents that are required for project closure needed to have been submitted for archiving and signed off on by all key stakeholders. I received the necessary sign-off on the project closure documents as of July 31, 2024, which allowed for the formal closing of the project.

Justification of Proposed versus Actual Project Outcomes

The Starlink project will possibly be perceived as one of the most accurate projects I have had the pleasure of collaborating on. So much of what was proposed was the actual project outcome. However, some things that were suspected did not end up happening. The risk register was plentiful; however, nearly none of the risks came to fruition. We did not face any natural disasters, SpaceX did not go out of business, none of our team members quit, none of our team members got injured, and we did not exceed the budget or the deadline. As exciting as it is, we finished the project before the deadline! The harsh weather that we encountered was not long lived and did not cause a delay long enough to have to cancel an installation for the day. We hired the entire team by April 15, 2024, as projected, trained them all by April 31, 2024, and had them in the field completing installations as of May 1, 2024. The project team successfully executed their two daily installations, each as planned. We had some call-ins during the project,

but I had already accounted for that by hiring two additional telecommunications technicians that we needed for the project to ensure adequate coverage in the event of someone being out. The Starlink project went very smoothly, and we ended it on July 31, 2024, instead of September 31, 2024, due to the excellent planning, stakeholder collaboration, highly effective communication, and transparency between stakeholders and the project team.

Lessons Learned

Lessons Learned Register			
Interpretation	**Item**	**Explanation**	**Action Items**
Win	Large Orders W/ Starlink	We learned that Starlink has had orders bigger than 600, and are more than capable of filling orders quickly that are of this size.	N/A
Win	Starlink Service Quality	Starlink performed better than predicted as a failover solution.	N/A
Issue	Time to Hire	Two weeks is not a good amount of time to hire as we hired the last hire on the very last day of the deadline.	Ensure all future project require 1 day allotted for planning activities.
Issue	Time for Meeting	Having meetings the day of installation is not advised as the first installations were rushed.	Ensure all future meetings have 1 day allotted for them to take place.

Although the project was a success, there were some events that didn't go as planned. I scheduled the sprint planning meeting on the same day as the technician's first two installations. This caused a rushed scenario but luckily, the installation process was not complex enough to lead to poor installation of the first ten Starlink hardware kits. To avoid this from happening in the future, I will ensure that meetings are grouped and allotted one day to be completed instead of overlapping them with events that should occur the day after the sessions. In addition to the meetings being held on the same day as the first installations, creating a rushed situation that could have led to rework in the project, I only allotted 14 days for 15 people to be hired and succeeded, but barely. I could have stretched the time for some of the events in the project as I allotted nearly thirty days for potential rework, which was excessive. I could have allowed some events that took a bit longer, such as hiring 15 project team members, a more reasonable amount of time. In the future, I will allot ten days past the minimum towards hiring new project team members instead of the exact time it would take. Though these issues did not have a critical impact on the project, they did cause dismay to the project team and stress to you as stakeholders who watched the timeline scrape by towards success. Those are not the emotions that I seek to instill. I hope that in this clarity, you see how the mishaps in this project have not only been identified but have been associated with a plan of action so that future projects do not have these issues.

The Starlink project had its fair share of issues that caused frustration. However, we experienced a few positive outcomes. Firstly, we received the entire Starlink order by the expected time, with all the parts functioning correctly. Secondly, Starlink's service performed much better than we had initially expected. One of the risks that I had originally posed was that Starlink might put us on a waitlist, and we would have to delay the entire execution process for an unknown amount of time. Luckily, that did not happen; when we reached out to Starlink, they advised that they had filled orders for over 1000 Starlink hardware kits in the past and that our request for 600 Starlink hardware kits would be easy. We received the Starlink equipment within the window that I had projected and successfully began installations on May 1, 2024. Lastly, I had proposed a risk that we may need a higher level of Starlink service, which would cost more money if the amount of traffic in our stores changed for some reason during the project. Once again, we were lucky in this situation as the analysis I had initially performed was 100% accurate regarding the stores that needed the lower-level Starlink business plan versus the locations that needed the premium Starlink business plan.

References

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